

Predicting Air-Pressure-System Failures in Scania Trucks

A Cost-Sensitive Machine Learning Approach By
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Introduction

Out on the road, Scania's big rigs depend on air pressure to stop and change gears. When that system falters, the truck might just quit moving risky business, plus expensive downtime follows.
From hidden signals in truck sensors, failure patterns begin to emerge. Built on real-world measurements stripped of identity, one glimpse into machine behavior takes shape. A challenge once posed by Scania reveals how rare breakdowns hide among routine operations. Where mistakes carry unequal consequences, detection becomes harder still. Majority cases drown out urgent warnings silence masks urgency. Prediction here means spotting what barely shows up..

Dataset

Data comes from Scania CV AB, later shared at the UCI Machine Learning Repository by Lindgren and Biteus in 2016. Out of every seven vehicles checked, six had gone through practice runs. One stood apart, part of a smaller group put to real use. Seventeen zero distinct numeric attributes show up here, seven of which represent pieces from a histogram. A spread like that often hides its original sources well.
Out of every hundred cases in training, just under two showed up positive. Meanwhile, testing saw about two and a third positives per hundred.
Empty spots marked "na" - some columns nearly gone, most cells blank. A string of fields where little remains past seventy percent wiped out.

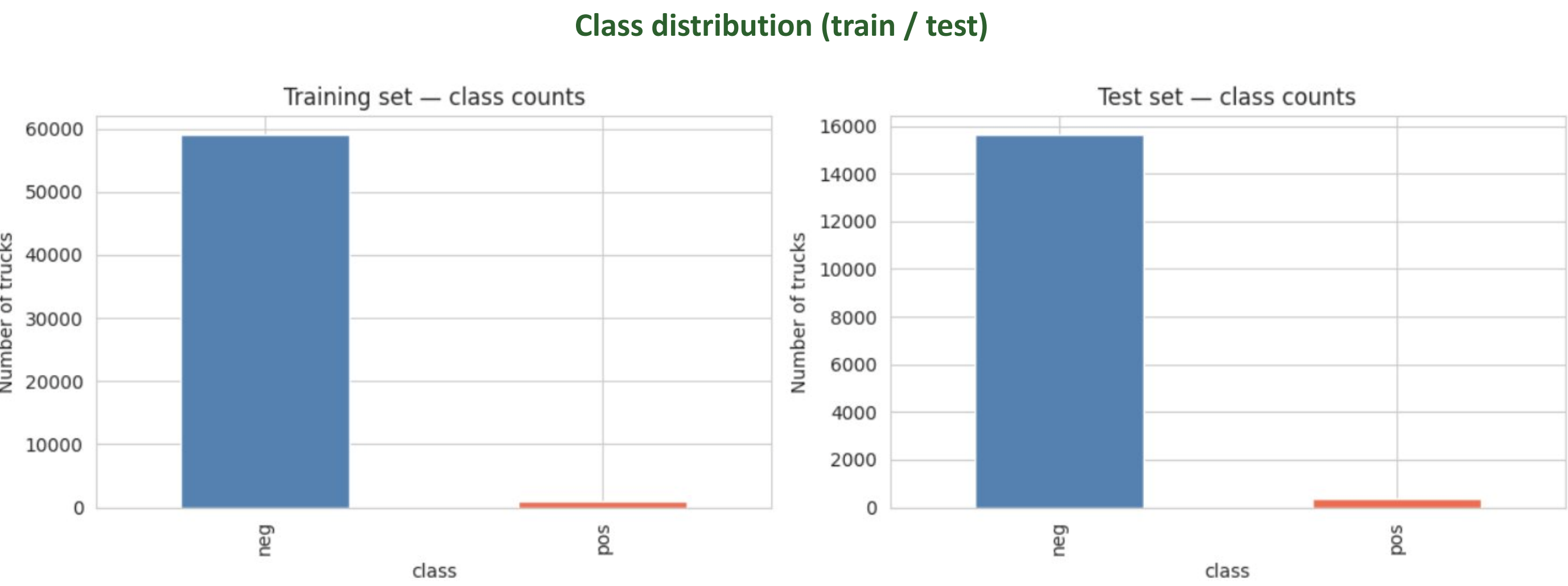


Figure 1. 1,000 of 60,000 training trucks (1.67%) are positives — severe imbalance.

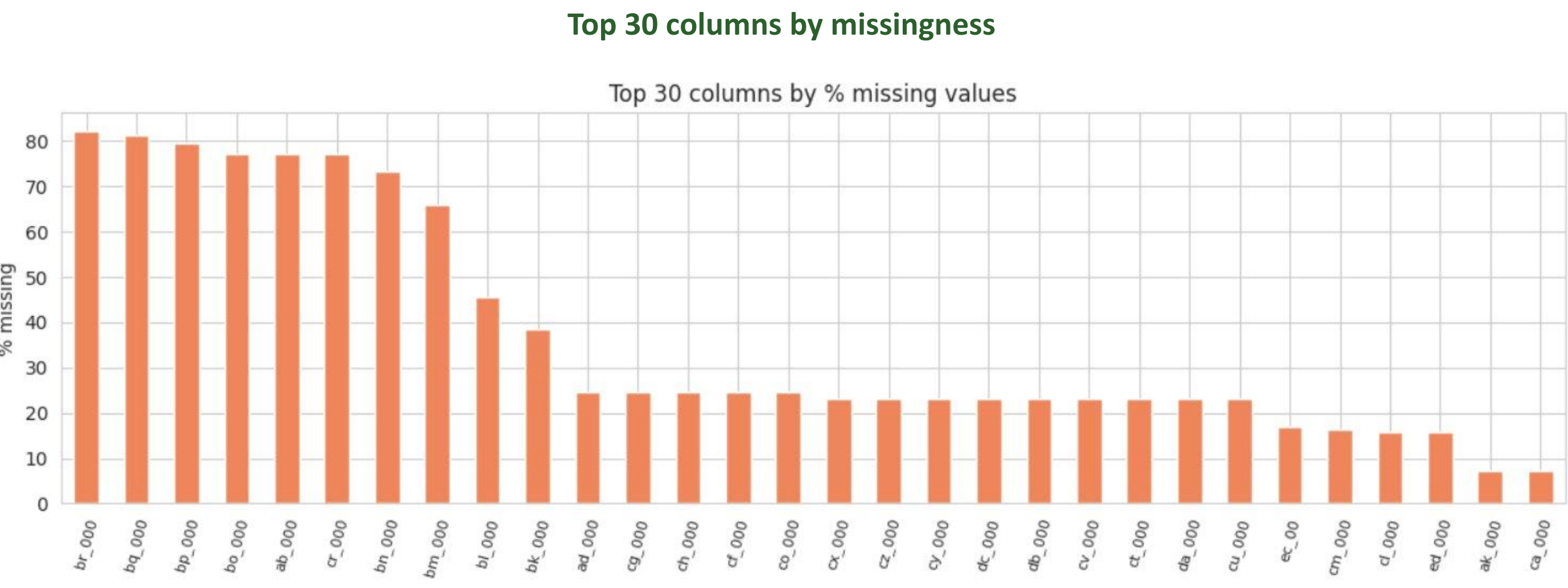


Figure 2. Seven columns exceed 70% missingness and were dropped, leaving 163 features.

Methodology

Pipeline - keeps training and testing separate, so nothing spills over
1.Drop 7 columns with > 70% missing values.
Filling leftover missing values using the middle number works well even when data leans one way.
3.Standard-scale all 163 features.
4.Train classifier with class_weight = "balanced".
Start by checking two hundred different points. From those, choose the one that keeps expenses lowest.

Cost Metric

$$\text{Total cost} = 10 \times \text{FP} + 500 \times \text{FN}$$

Missing a faulty truck (FN) costs 50× more than a needless inspection (FP).



Figure 3. Cost vs. decision threshold — minimum 10,410 at t = 0.061.

Models Compared

1.Logistic Regression(balanced)
2.Decision Tree(balanced)
3.K-Nearest Neighbors(k = 5)
4.Random Forest(balanced)
5.Gradient Boosting
6.XGBoost(scale_pos_weight = 59)
Out of all models tested, Random Forest stood out when price mattered most. The setup got tweaked using a three-part split method that focused on ROC scores - highest score reached 0.9874.

Results

Total cost: 10,410

Random Forest (grid-tuned) threshold 0.061
ROC-AUC = 0.9938 F1 (pos) = 0.5919 FN = 11

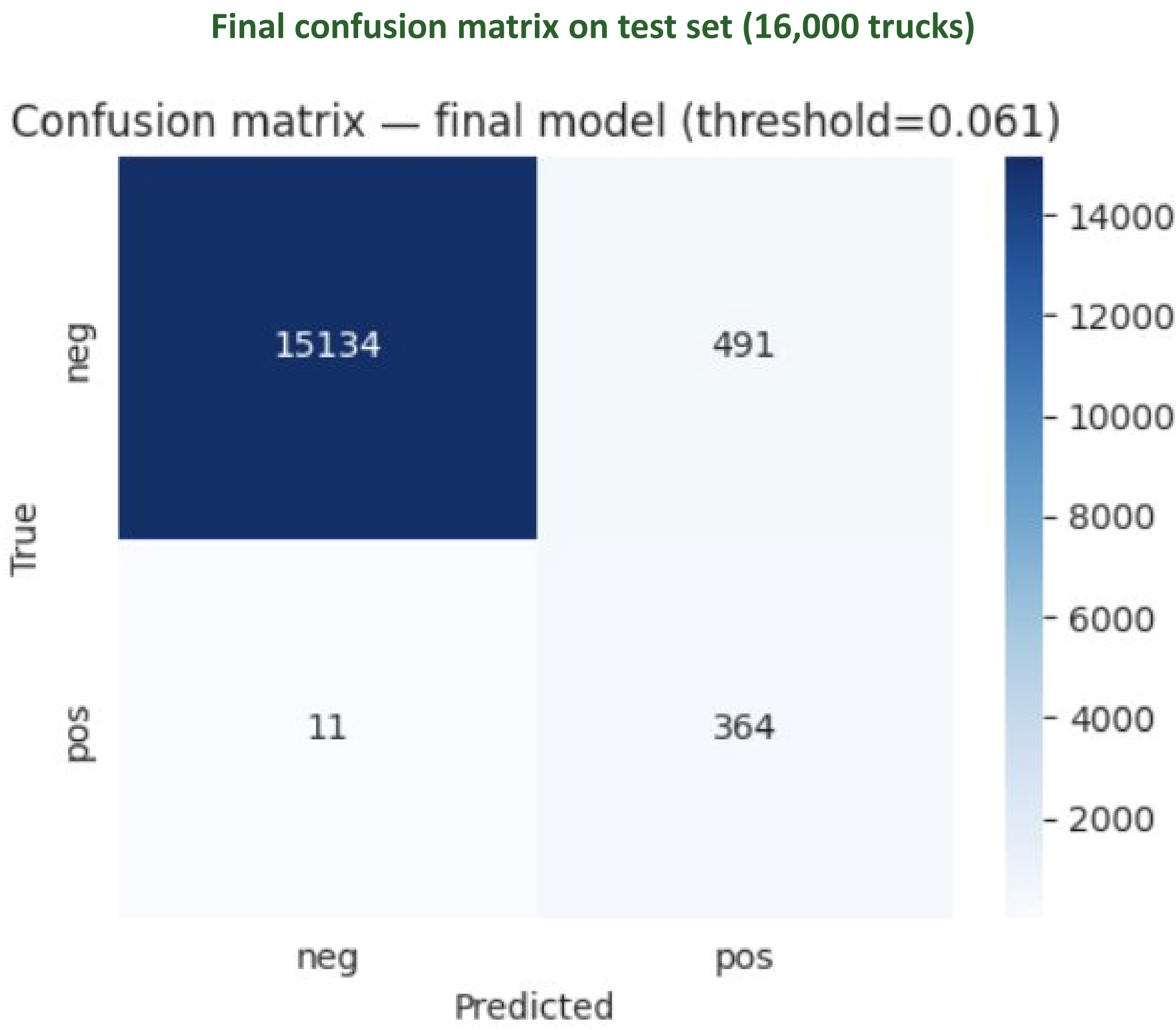
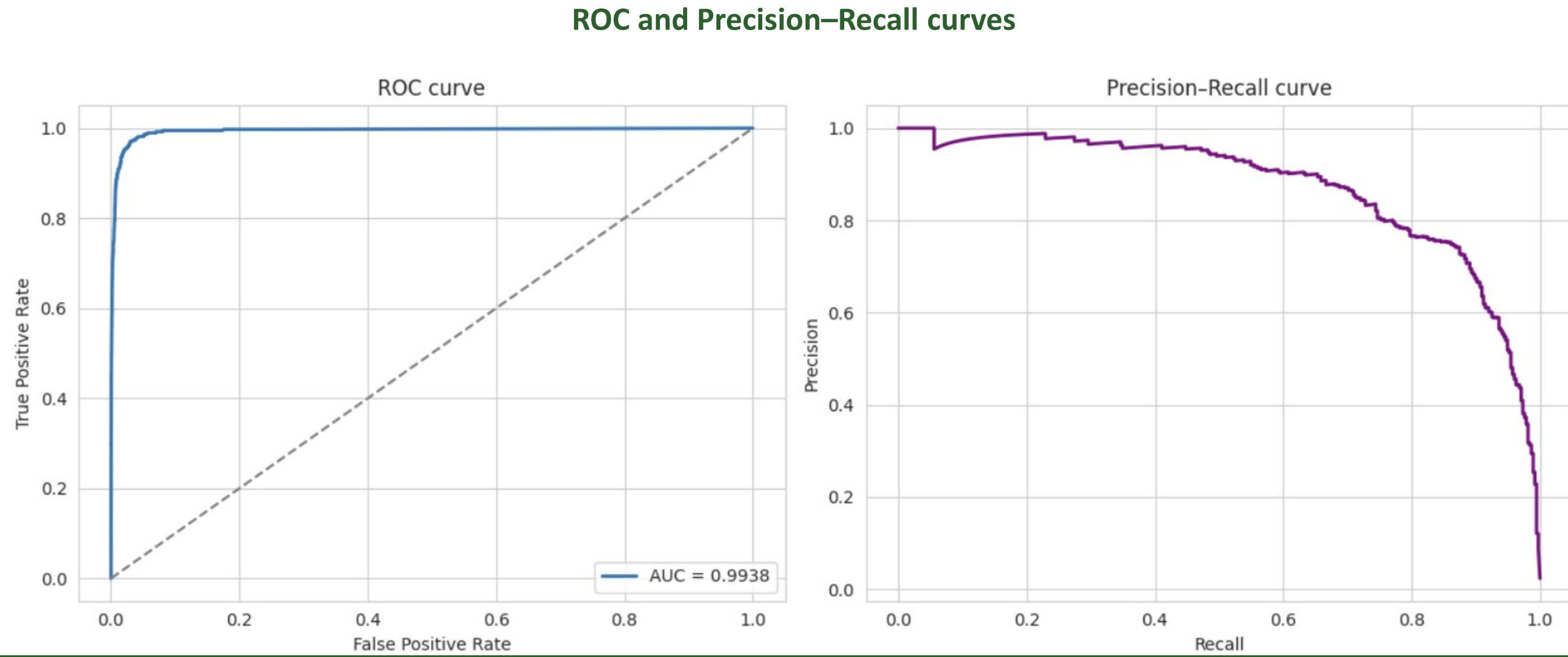


Figure 4. Only 11 of 375 faulty trucks missed, 491 false alarms.



Conclusions

Out of nowhere, Random Forest using class_weight="balanced" outperformed Logistic Regression, KNN, and Decision Tree costing up to eight times less overall. While those models struggled, the edge went solidly to Random Forest, leaving others behind by at least 40 percent in efficiency.
Costs start rising past 60,000 when sticking to the usual 0.5 cutoff - tuning matters a lot. That sweet spot? It lands at 0.061 instead. Three times higher price tag after using SMOTE 28,700 instead of 9,100. Weighting classes by importance worked better on its own.

References

- [1] Lindgren & Biteus. APS Failure at Scania Trucks. UCI ML Repository, 2016. <https://archive.ics.uci.edu/dataset/421/aps+failure+at+scania+trucks>
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- [3] Breiman, L. Random Forests. Machine Learning, 45(1), 5–32, 2001. <https://link.springer.com/article/10.1023/A:1010933404324>